THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 15°	Angles $a, a+d, a+2d, a+3d$ add up to 360°.
2	**(d)** 70 cm	Sides 15 cm and 20 cm.
3	**(c)** 72 cm	Side $= 18$ cm.
4	**(a)** 90 cm	Half-diagonals and the side form a right triangle: half of the other diagonal $= \sqrt{51^2 - 24^2} = 45$.
5	**(c)** 1344 cm^2	Area $= \frac{1}{2} d_1 d_2$.
6	**(b)** 168 cm^2	Other side $= 24$ cm; area $= 7 \times 24$.
7	**(d)** 87	Opposite angles are equal.
8	**(a)** 131 cm	Each side of the new figure is half a diagonal; there are two of each.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$2 \times 180^\circ = 360^\circ$; R is the standard proof.
10	**(c)** A is true but R is false.	A is a property of parallelograms; R is false (the diagonals of a non-rectangular parallelogram are unequal).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Half of the given diagonal $= 24$ cm; Half of the other diagonal $= \sqrt{51^2 - 24^2} = 45$ cm, so the diagonal is 90 cm (1); Area $= \frac{1}{2} \times 48 \times 90 = 2160$ cm^2; Height $= \frac{\text{area}}{\text{side}} = \frac{2160}{51} = 42.35$ cm (1).	1+1
12	Let the angles be $a, a+40, a+80, a+120$; $4a + 240 = 360$ gives $a = 30$ (1); Angles: $30^\circ, 70^\circ, 110^\circ, 150^\circ$; $30^\circ + 150^\circ = 180^\circ$ (and $70^\circ + 110^\circ$ also 180°), so they are supplementary (1).	1+1
13	Opposite sides are equal: $AB = CD$; $2x + 1 = x + 23$ gives $x = 22$ (1); $AB = CD = 45$ cm and $BC = AD = 24$ cm; Perimeter $= 2(45 + 24) = 138$ cm (1).	1+1
14	Half of the given diagonal $= 7$ cm; Half of the other diagonal $= \sqrt{25^2 - 7^2} = 24$ cm, so the diagonal is 48 cm (1); Area $= \frac{1}{2} \times 14 \times 48 = 336$ cm^2; Height $= \frac{\text{area}}{\text{side}} = \frac{336}{25} = 13.44$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Diagonals bisect each other, so $OA = OC$; $5x + 3 = x + 147$ gives $x = 36$ (1); $OA = 183$ cm (1); $AC = 2 \times 183 = 366$ cm (1).	1+1+1
16	$\angle B = 180^\circ - 88^\circ = 92^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 44^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 46^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 44^\circ - 46^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
17	$AD = 18$ cm (D is the midpoint); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 18$ cm (1); Hence $AE = EC$ (1).	1+1+1
	OR: Diagonals bisect each other, so $OA = OC$; $4x + 3 = 2x + 57$ gives $x = 27$ (1); $OA = 111$ cm (1); $AC = 2 \times 111 = 222$ cm (1).	1+1+1
18	$\angle B = 180^\circ - 142^\circ = 38^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 71^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 19^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 71^\circ - 19^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
	OR: $AD = 16$ cm (D is the midpoint); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 9$ cm (1); Hence $AE = EC$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 360° (1). (ii) $2 + 3 + 4 + 6 = 15$ parts, so one part $= \frac{360^\circ}{15} = 24^\circ$ (1). (iii) $6 \times 24^\circ = 144^\circ$ (1). (iv) The angles are $48^\circ, 72^\circ, 96^\circ, 144^\circ$ (1); a parallelogram has equal opposite angles, but opposite angles such as 48° and 96° differ, so it is not a parallelogram (1).	1+1+1+2

20	Each side of the new figure is half a diagonal of the rectangle (midpoint theorem), and the diagonals of a rectangle are equal, so all four sides are equal (1); Diagonal of the rectangle $= \sqrt{1764 + 20736} = 150$ cm, so each side $= 75$ cm (1); Perimeter $= 4 \times 75 = 300$ cm (1); Its diagonals are the sides of the rectangle, so area $= \frac{1}{2} \times 42 \times 144 = 3024$ cm^2 (2).	1+1+1+2
	OR: Diagonals of a parallelogram bisect each other, so $OA = OC$ and $OB = OD$ (1); $OA^2 + OB^2 = 9 + 16 = 25 = AB^2$ (1); By the converse of the Baudhāyana–Pythagoras theorem, $\angle AOB = 90^\circ$, so the diagonals are perpendicular (1); A parallelogram with perpendicular diagonals is a rhombus, so all sides equal (1); Perimeter $= 4 \times 5 = 20$ cm (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 28 cm	Side $= 7$ cm.
2	**(b)** 48 cm	Half-diagonals and the side form a right triangle: half of the other diagonal $= \sqrt{25^2 - 7^2} = 24$.
3	**(b)** 3024 cm^2	Area $= \frac{1}{2} d_1 d_2$.
4	**(d)** 3240 cm^2	Other side $= 120$ cm; area $= 27 \times 120$.
5	**(c)** 101	Opposite angles are equal.
6	**(d)** 202 cm	Each side of the new figure is half a diagonal; there are two of each.
7	**(d)** 60°	Angles $a, a+d, a+2d, a+3d$ add up to 360°.
8	**(c)** 112 cm	Sides 24 cm and 32 cm.
9	**(d)** A is false but R is true.	A is false (the diagonals of a rhombus are equal only if it is a square); R is true.
10	**(b)** Both A and R are true but R is not the correct explanation of A.	A is true because the opposite sides of a rectangle are parallel; R is true but does not explain A.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Let the angles be $a, a+10, a+20, a+30$; $4a + 60 = 360$ gives $a = 75$ (1); Angles: $75^\circ, 85^\circ, 95^\circ, 105^\circ$; $75^\circ + 105^\circ = 180^\circ$ (and $85^\circ + 95^\circ$ also 180°), so they are supplementary (1).	1+1
12	Opposite sides are equal: $AB = CD$; $2x + 5 = x + 21$ gives $x = 16$ (1); $AB = CD = 37$ cm and $BC = AD = 20$ cm; Perimeter $= 2(37 + 20) = 114$ cm (1).	1+1
13	Half of the given diagonal $= 21$ cm; Half of the other diagonal $= \sqrt{75^2 - 21^2} = 72$ cm, so the diagonal is 144 cm (1); Area $= \frac{1}{2} \times 42 \times 144 = 3024$ cm^2; Height $= \frac{\text{area}}{\text{side}} = \frac{3024}{75} = 40.32$ cm (1).	1+1
14	Let the angles be $a, a+14, a+28, a+42$; $4a + 84 = 360$ gives $a = 69$ (1); Angles: $69^\circ, 83^\circ, 97^\circ, 111^\circ$; $69^\circ + 111^\circ = 180^\circ$ (and $83^\circ + 97^\circ$ also 180°), so they are supplementary (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Diagonals bisect each other, so $OA = OC$; $5x + 5 = x + 77$ gives $x = 18$ (1); $OA = 95$ cm (1); $AC = 2 \times 95 = 190$ cm (1).	1+1+1
16	$\angle B = 180^\circ - 78^\circ = 102^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 39^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 51^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 39^\circ - 51^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
17	$AD = 14$ cm (D is the midpoint); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2} BC = 9$ cm (1); Hence $AE = EC$ (1).	1+1+1
	OR: Diagonals bisect each other, so $OA = OC$; $5x + 5 = 3x + 59$ gives $x = 27$ (1); $OA = 140$ cm (1); $AC = 2 \times 140 = 280$ cm (1).	1+1+1
18	$\angle B = 180^\circ - 138^\circ = 42^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 69^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 21^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 69^\circ - 21^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
	OR: $AD = 12$ cm (D is the midpoint); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2} BC = 20$ cm (1); Hence $AE = EC$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Opposite angles are equal, so $\angle R = 65^\circ$ (1). (ii) $\angle Q = 180^\circ - 65^\circ = 115^\circ$ (1). (iii) $2(8 + 5) = 26$ cm (1). (iv) $OP = 7$ cm and $OQ = 5$ cm (1); the diagonals of a parallelogram bisect each other (1).	1+1+1+2

20	Each side of the new figure is half a diagonal of the rectangle (midpoint theorem), and the diagonals of a rectangle are equal, so all four sides are equal (1); Diagonal of the rectangle $= \sqrt{900 + 5184} = 78$ cm, so each side $= 39$ cm (1); Perimeter $= 4 \times 39 = 156$ cm (1); Its diagonals are the sides of the rectangle, so area $= \frac{1}{2} \times 30 \times 72 = 1080$ cm ^{2} (2).	1+1+1+2
	OR: Diagonals of a parallelogram bisect each other, so $OA = OC$ and $OB = OD$ (1); $OA^2 + OB^2 = 576 + 2025 = 2601 = AB^2$ (1); By the converse of the Baudhāyana–Pythagoras theorem, $\angle AOB = 90^\circ$, so the diagonals are perpendicular (1); A parallelogram with perpendicular diagonals is a rhombus, so all sides equal (1); Perimeter $=$ $4 \times 51 = 204$ cm (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Quadrilaterals** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 720 cm^2	Area $= \frac{1}{2}d_1d_2$.
2	**(a)** 420 cm^2	Other side $= 21$ cm; area $= 20 \times 21$.
3	**(b)** 66	Opposite angles are equal.
4	**(b)** 207 cm	Each side of the new figure is half a diagonal; there are two of each.
5	**(d)** 57°	Angles $a, a+d, a+2d, a+3d$ add up to 360°.
6	**(b)** 56 cm	Sides 12 cm and 16 cm.
7	**(d)** 48 cm	Side $= 12$ cm.
8	**(a)** 80 cm	Half-diagonals and the side form a right triangle: half of the other diagonal $= \sqrt{41^2 - 9^2} = 40$.
9	**(a)** Both A and R are true and R is the correct explanation of A.	$DE = \frac{1}{2}BC = 5$ cm, directly from the midpoint theorem.
10	**(a)** Both A and R are true and R is the correct explanation of A.	$180^\circ - 70^\circ = 110^\circ$, using R.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Opposite sides are equal: $AB = CD$; $2x + 8 = x + 35$ gives $x = 27$ (1); $AB = CD = 62$ cm and $BC = AD = 28$ cm; Perimeter $= 2(62 + 28) = 180$ cm (1).	1+1
12	Half of the given diagonal $= 9$ cm; Half of the other diagonal $= \sqrt{41^2 - 9^2} = 40$ cm, so the diagonal is 80 cm (1); Area $= \frac{1}{2} \times 18 \times 80 = 720$ cm^2; Height $= \frac{\text{area}}{\text{side}} = \frac{720}{41} = 17.56$ cm (1).	1+1
13	Let the angles be $a, a+22, a+44, a+66$; $4a + 132 = 360$ gives $a = 57$ (1); Angles: $57^\circ, 79^\circ, 101^\circ, 123^\circ$; $57^\circ + 123^\circ = 180^\circ$ (and $79^\circ + 101^\circ$ also 180°), so they are supplementary (1).	1+1
14	Opposite sides are equal: $AB = CD$; $3x + 3 = x + 67$ gives $x = 32$ (1); $AB = CD = 99$ cm and $BC = AD = 38$ cm; Perimeter $= 2(99 + 38) = 274$ cm (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Diagonals bisect each other, so $OA = OC$; $4x + 8 = 3x + 36$ gives $x = 28$ (1); $OA = 120$ cm (1); $AC = 2 \times 120 = 240$ cm (1).	1+1+1
16	$\angle B = 180^\circ - 110^\circ = 70^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 55^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 35^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 55^\circ - 35^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
17	$AD = 9$ cm (D is the midpoint); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 7$ cm (1); Hence $AE = EC$ (1).	1+1+1
	OR: Diagonals bisect each other, so $OA = OC$; $5x + 1 = 3x + 35$ gives $x = 17$ (1); $OA = 86$ cm (1); $AC = 2 \times 86 = 172$ cm (1).	1+1+1
18	$\angle B = 180^\circ - 128^\circ = 52^\circ$; Diagonal AC bisects $\angle A$: $\angle OAB = 64^\circ$ (1); $\angle OBA = \frac{1}{2}\angle B = 26^\circ$ (1); In $\triangle AOB$: $\angle AOB = 180^\circ - 64^\circ - 26^\circ = 90^\circ$ (diagonals are perpendicular) (1).	1+1+1
	OR: $AD = 13$ cm (D is the midpoint); By the converse of the midpoint theorem, E is the midpoint of AC (1); $DE = \frac{1}{2}BC = 13$ cm (1); Hence $AE = EC$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $DE = \frac{1}{2} \times 24 = 12$ m (1). (ii) DE is parallel to BC (1). (iii) $AD = 9$ m, $AE = 13$ m (1). (iv) Perimeter of $\triangle ADE = 9 + 13 + 12 = 34$ m (1); perimeter of $\triangle ABC = 18 + 24 + 26 = 68$ m, so it is half (1).	1+1+1+2

20	Each side of the new figure is half a diagonal of the rectangle (midpoint theorem), and the diagonals of a rectangle are equal, so all four sides are equal (1); Diagonal of the rectangle $= \sqrt{1600 + 1764} = 58$ cm, so each side $= 29$ cm (1); Perimeter $= 4 \times 29 = 116$ cm (1); Its diagonals are the sides of the rectangle, so area $= \frac{1}{2} \times 40 \times 42 = 840$ cm^2 (2).	1+1+1+2
	OR: Diagonals of a parallelogram bisect each other, so $OA = OC$ and $OB = OD$ (1); $OA^2 + OB^2 = 81 + 144 = 225 = AB^2$ (1); By the converse of the Baudhāyana–Pythagoras theorem, $\angle AOB = 90^\circ$, so the diagonals are perpendicular (1); A parallelogram with perpendicular diagonals is a rhombus, so all sides equal (1); Perimeter $= 4 \times 15 = 60$ cm (1).	1+1+1+1+1

Mistakes to avoid:

- Saying the diagonals of every parallelogram are equal; that holds only for a rectangle (and square).
- Saying the diagonals of a rhombus are equal; they are perpendicular bisectors of each other, equal only in a square.
- Taking adjacent angles of a parallelogram to be equal; they add up to 180° and the *opposite* angles are equal.
- Using the full diagonals instead of half-diagonals when finding the side of a rhombus with the Pythagoras theorem.
- In the midpoint theorem, writing $DE = 2BC$ or $DE = \frac{1}{2}AB$; the segment is half the *third* side that it is parallel to.
- Forgetting that a quadrilateral's angles must add up to 360° (not 180°) when solving for x.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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